MMSE Pair Scheduling: Computational Complexity and Certified Power Allocation

Pooya Hejazi and Masoud Sabaei

***Abstract*—Uplink pair scheduling couples pairing, resource assignment and transmit powers when a minimum mean-square error (MMSE) receiver is recomputed after each power change. We study a common receiver serving every user once, with two users per orthogonal resource, positive bounded powers and a shared budget. We prove fixed-power ideal-threshold attainment NP-complete for three receive dimensions and rational unit channels from a fixed finite alphabet. A continuous-power extension retains a constant additive separation. We then derive the sharp whole-quadrant concavity condition for the pair log-rate and characterize rectangular power pricing by at most seventeen candidates. These results support an offline reference that returns a feasible allocation and a verified upper bound on the continuous optimum. Following a disclosed channel-window correction, we compare methods on 24 eight-user and eight sixteen-user networks in two ray-traced scenes. Pricing tightens 11 of 24 main bounds and all eight scale bounds, but adds neither achieved rate over curvature nor a 0.01-nat target certificate. The registered certification-fraction difference is zero, with conservative 95% interval $[-0.3449, 0.3449]$. Thus valid bounds quantify nominal allocation quality, while useful target certification and scalable near-optimality remain unestablished.**

***Index Terms*—MMSE pair scheduling, computational complexity, certified power allocation.**

I. Introduction

UPLINK pair scheduling lets two users share a resource while a linear receiver separates their signals without successive interference cancellation (SIC). Their rates depend on channel alignment and powers. Changing a power also changes the minimum mean-square error (MMSE) filter, coupling pairing and power allocation.

The two-user no-SIC rate is established [1]. Fan et al. [2] optimize MMSE-aware pairing at prescribed powers; Ruder et al. [3] also examine imperfect scheduling information. We ask how the quality of a joint schedule and continuous powers can be bounded for a declared channel realization. We consider a common receiver, two users per orthogonal resource, service of every user once, positive bounded powers and one shared budget.

The authors are with the Department of Computer Engineering, Amirkabir University of Technology, Tehran, Iran (e-mail: pooya.hejazi@aut.ac.ir; sabaei@aut.ac.ir). Corresponding author: Masoud Sabaei. ORCID: Pooya Hejazi, 0000-0001-5877-5368; Masoud Sabaei, 0000-0001-8024-8898.

A feasible schedule gives an achievable rate but not its distance from the optimum. A finite power grid can miss better continuous allocations. We therefore seek both a feasible allocation and a valid continuous upper bound; their difference bounds the unknown loss from optimality. We make three contributions:

1) We prove fixed-power ideal-threshold attainment NP-complete with three receive dimensions and rational unit channels from a fixed finite alphabet. A continuous-power extension retains a constant additive separation.
2) We derive the sharp whole-quadrant concavity condition for the pair log-rate and characterize rectangular pricing by at most seventeen candidates, with an interior polynomial of degree at most five.
3) We construct independently checkable global bounds and compare grid, curvature, pricing and matching–WMMSE methods under common constraints and budgets. Pricing tightens some bounds but adds no rate gain over curvature or superiority in the registered target-certification comparison.

The evaluation uses one eight-antenna receiver in each of two ray-traced scenes. No main or scale case reaches the 0.01-nat target. An original channel-window error and its separately registered correction are disclosed with both sets of outcomes. The supplement supplies full proofs, comparisons and experimental records. Throughout, $\ln$ is the natural logarithm and $(\cdot)^H$ the conjugate transpose.

II. Related Work

A. Pair scheduling and computational complexity

Table I compares the closest receiver, scheduling and optimization precedents. Ruder et al. [4] also distinguish homogeneous-resource pairing from frequency-selective grouping. These works establish the rate and allocation architecture; our question concerns continuous powers and a bound for an integral full-service schedule.

Prior hardness results address distinct allocation models. Prasad et al. [5] study LTE-A multistream scheduling with contiguity and control constraints; Salaün et al. [6] study subcarrier/power allocation with SIC and

individual budgets. Mollanoori and Ghaderi [7] consider full-service scalar SIC scheduling with variable group sizes. Razaviyayn et al. [8] establish three-antenna interference-alignment hardness, and Liu [9] establishes two-antenna continuous-transceiver hardness. Our theorem concerns their different conjunction with a common no-SIC receiver, exactly two users per resource and a finite rational channel alphabet.

B. Power optimization and bounds

Shen and Yu's quadratic transform [10] gives an exact auxiliary reformulation with stationary-point convergence under its assumptions. Their matching-based scheduling [11] and Wang et al.'s weighted-MMSE precoding with error covariance [12] motivate our common-model comparator. The fixed-filter and joint optimization guarantees remain distinct, as Table I explains.

Ul Hassan et al. [13] derive finite candidates for two-link power sharing with fixed interference gains and an exhausted sum budget. Sharma and Teneketzis [14] obtain decentralized guarantees under concave utilities; their MMSE example uses negative MSE rather than our log-rate. We specialize mixed-monotone optimization [15] to the recomputed pair rate. Barnhart et al.'s branch-and-price framework [16] requires complete or bounded pricing. Zhao and Pottie's divisible-spectrum convexification [17] permits sharing unavailable to one integral schedule.

C. Recent resource-allocation designs

Ou et al. [21] transmit two streams per dual-polarized user in a no-SIC uplink, with estimation errors and a finite-blocklength dispersion penalty. Algorithm 1 fixes combiners before the power loop. Our single-stream users, integral pair/resource decisions, shared budget and recomputed-MMSE Shannon rate define a different problem.

Chen et al. [22] jointly design downlink grouping, subcarriers and hybrid beamforming under beam squint; at-least-once service and changing penalties differ from our model and do not ensure sum-rate monotonicity. Elyasi and Vosoughi [23] combine multicell MMSE, pilots and powers, using a high-SINR approximation and frozen coefficients for sum SE. Daei et al. [24] use heuristic frame search and projected power ascent under channel aging; their fixed-point convergence result concerns SE evaluation.

Fu et al. [25] optimize downlink power and regularization after large-scale-fading association using a large-system deterministic equivalent and a reported stationarity guarantee, rather than a finite-instance bound. Liu et al. [26] develop permutation-aware mobile-AP learning; permutation guarantees do not establish global reward optimality. Mobini et al. [27] combine heuristic fronthaul-aware association and fixed-association stationarity with MMSE-SIC and per-AP budgets. Their architectural and information assumptions preclude treating reported gains as matched baselines here. The supplement gives detailed comparisons and reading-version boundaries.

III. System Model and Problem Formulation

A. Uplink transmission and information assumptions

We consider $K = 2S$ single-antenna users and a common M-dimensional receiver over S orthogonal resources. Exactly two users share each resource, and every user is served once. This full-service requirement separates the pairing decision from admission control: a user with an unfavorable channel cannot be dropped to improve the objective.

Let $\mathcal{K} = \{1, \dots, K\}$, $\mathcal{S} = \{1, \dots, S\}$, and $\mathcal{E} = \{\{u, v\} : u < v,\ u, v \in \mathcal{K}\}$. The channel of user i on resource s is $\mathbf{h}_{is} \in \mathbb{C}^M$. After whitening by the known positive-definite noise covariance, the received signal for pair $e = \{u, v\}$ is

$$\mathbf{y}_s = \sqrt{p_u}\mathbf{h}_{us}a_u + \sqrt{p_v}\mathbf{h}_{vs}a_v + \mathbf{n}_s, \qquad \mathbb{E}[\mathbf{n}_s\mathbf{n}_s^H] = \mathbf{I}_M. \tag{1}$$

Independent, zero-mean, unit-variance complex Gaussian symbols are decoded with residual interference treated as noise; finite constellations and decoding errors are not modeled. Powers are in watts. For scalar noise, (1) divides the physical channel by the noise standard deviation, without reapplying path loss or noise normalization.

The scheduler and receiver know the same channel realization. We use nominal to mean that optimization and certification refer to this specified realization, including its stored precision. A guarantee under imperfect channel state information (CSI) would additionally require an uncertainty model.

Users satisfy $0 < p_{\min} \le p_i \le p_{\max}$ and a shared budget $\sum_i p_i \le P$. The positive lower bound ensures that service cannot be removed through zero power. Since every pair is eligible, $Kp_{\min} \le P$ guarantees a feasible power allocation for any schedule. Equal-bandwidth resources are mutually orthogonal. We maximize unweighted sum rate without quality-of-service constraints, SIC or additional local power budgets.

Fig. 1 places the full-service architecture in an illustrative physical setting. The evaluated receiver is one eight-antenna AP; the theory allows general M. Pair identities and resource labels are optimization decisions, not fixed sequential-index pairing.

TABLE I
CLOSEST STUDIES, SELECTED BY RECEIVER, ALLOCATION MODEL AND OPTIMIZATION GUARANTEE. LIMITATIONS DESCRIBE SCOPE OF APPLICABILITY, NOT MEASURED INFERIORITY.

Work	Contribution and strength	Relevant scope limitation	Question addressed here
Haustein [1]	Derives recomputed-MMSE pair rates and identifies nonconcavity	Receiver analysis does not settle joint schedule/power quality	Restricted hardness, sharp curvature and continuous bounds
Fan et al. [2]	MMSE-aware pairing and binary frequency allocation with branch-and-bound	Prescribed powers and contiguous resource patterns	Positive continuous powers with a shared budget
Ruder et al. [3]	Pairing with predicted/interpolated scheduling CSI and pilot accounting	Fixed powers; perfect payload-equalization CSI	Nominal continuous-power certification, without a mismatch guarantee
Mostafa et al. [18]	Channel-orthogonality reduction and pair/master offloading design	Growing dimension; SIC, admission and delay constraints	Fixed three-dimensional finite-alphabet threshold hardness
Shi et al. [19]	Rate–MMSE relation and globally solved fixed-filter power steps	Alternating joint problem remains nonconvex	Complete pair pricing with the receiver recomputed
Viterbo–Hottinen [20]	Equal-power MMSE pairing; distinguishes disjoint pairs and expanded cyclic schedules	Resource-invariant channels and different scheduling families	Resource-labelled full-service pairs with continuous powers
Shen–Yu [11]	Matching-based joint scheduling, power and beamformer updates	Continuous stationarity for a fixed schedule is not joint global optimality	Verified bounds for an integral common-receiver allocation
Wang et al. [12]	Weighted-MMSE precoding with error covariance and centralized/distributed processing	Statistical/estimated-CSI objective and iterative precoding	Fixed-channel scalar powers and a common-model comparator
Matthiesen et al. [15]	Globally convergent mixed-monotone branch-reduce-bound	Subdivision may be costly; bounds depend on the objective	Pair-specific curvature and pricing within that framework
Ou et al. [21]	SIC-free dual-polarized uplink with estimated CSI and finite-blocklength power control	Fixed power-independent combiners; per-user budgets; conditional power stationarity	Integral pair/resource decisions and bounds with recomputed MMSE

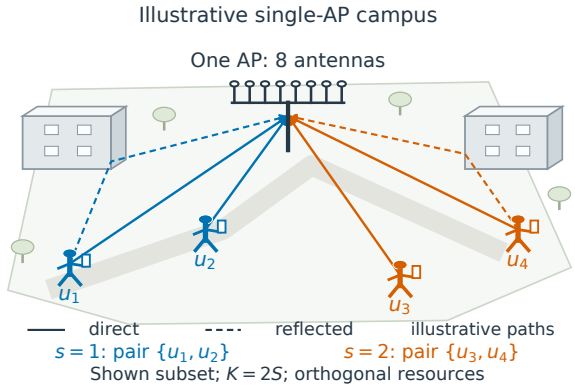


Fig. 1. Illustrative uplink: one eight-antenna AP, pairs on orthogonal resources and linear MMSE without SIC. Each user is served once under positive power bounds and a shared budget. Geometry and propagation paths are schematic, not measurements.

B. Recomputed linear MMSE reception

For each selected pair, the receiver recomputes the linear MMSE filter at the selected powers and treats the other user's signal as noise. We use the established two-user rate derived by Haustein [1]; our analysis concerns its curvature and its role in joint allocation. Define

$$g_{is} = \|\mathbf{h}_{is}\|^2 \geq 0,$$
$$r_{uvs} = \frac{|\mathbf{h}_{us}^H \mathbf{h}_{vs}|^2}{g_{us} g_{vs}} \in [0,1] \quad (g_{us} g_{vs} > 0). \tag{2}$$

Here r_{uvs} is squared normalized channel correlation: zero denotes orthogonal channels and one denotes collinear nonzero channels. If either gain is zero, set $r_{uvs} = 0$; the corresponding zero normalized power makes this convention consistent with the MMSE rates. The zero-channel user still receives service and minimum power. With $x = g_{us} p_u$ and $y = g_{vs} p_v$, the matrix inversion lemma gives the signal-to-interference-plus-noise ratios (SINRs)

$$\gamma_u = x \frac{1 + (1 - r_{uvs}) y}{1 + y}, \qquad \gamma_v = y \frac{1 + (1 - r_{uvs}) x}{1 + x}. \tag{3}$$

Consequently the pair sum rate, expressed in natural-log units per channel use, is

$$q_r(x, y) = 2 \ln d - \ln(1 + x) - \ln(1 + y),$$
$$d = 1 + x + y + (1 - r) xy. \tag{4}$$

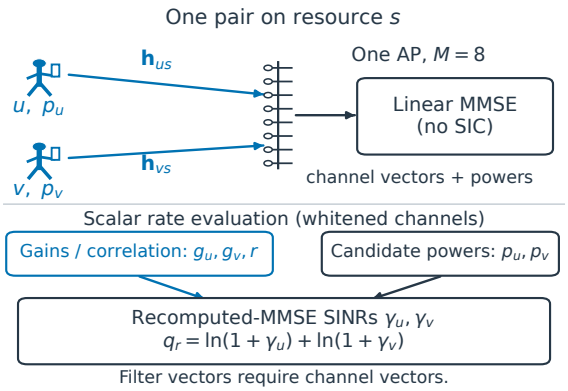


Fig. 2. One pair on a common resource. Whitened gains, correlation and powers determine (3)–(4); constructing receive filters requires full channel vectors. The schematic adds no measured geometry, channel-estimation or SIC model.

We write $q_{uvs}(p_u, p_v) = q_{r_{uvs}}(g_{us}p_u, g_{vs}p_v)$. Formula (4) includes receiver recomputation; fixing receiver coefficients while varying powers gives a different objective.

The reported sum of per-resource spectral efficiencies (SEs) is

$$Q = \frac{1}{\ln 2} \sum_{s=1}^{S} q_{e_s,s}. \tag{5}$$

For resource bandwidth B_r and data fraction η, throughput is $T = \eta B_\mathrm{r} Q$ bit/s, and efficiency normalized by the total occupied resource bandwidth is $\eta Q/S$. Neither B_r nor η follows from a channel array alone. We therefore distinguish Q, radiated power and computational time from system energy efficiency, which would require an amplifier, circuit and processing-power model.

Fig. 2 shows the scalar rate calculation for one pair. Gains and correlation suffice to evaluate the SINRs with MMSE recomputation through (3); constructing the receive-filter vectors themselves requires the channel vectors.

C. Joint continuous problem

Let $z_{uvs} \in \{0, 1\}$ indicate the pair placed on resource s. The joint problem is

$$\begin{aligned}
\mathcal{P}_\mathrm{C}: \quad \max_{z,p} \quad & \sum_{s\in\mathcal{S}} \sum_{u<v} z_{uvs} q_{uvs}(p_u, p_v) \\
\text{s.t.} \quad & \sum_{s} \Big(\sum_{v<i} z_{vis} + \sum_{v>i} z_{ivs} \Big) = 1 \ (i \in \mathcal{K}), \\
& \sum_{u<v} z_{uvs} = 1 \quad (s \in \mathcal{S}), \\
& p_\mathrm{min} \le p_i \le p_\mathrm{max} \quad (i \in \mathcal{K}), \\
& \sum_i p_i \le P
\end{aligned} \tag{6}$$

The first two equalities enforce service once per user and one pair per resource. Powers can be indexed by users because each appears once, but its rate still depends on its partner and resource. Problem (6) therefore combines discrete scheduling and continuous powers. Even concave pair utilities would leave the scheduling decisions to be resolved.

Viterbo and Hottinen [20] distinguish fixed-power disjoint pairing from an expanded cyclic scheduling family. For our disjoint pairs with resource-invariant channels, the weights q_{uvs} do not depend on s, and maximum-weight perfect matching solves the scheduling step. Edmonds' general-graph matching framework [28] applies; an unrestricted bipartite assignment need not produce disjoint pairs. Resource-dependent channels require the resource labels to remain in the problem.

D. Finite-grid integer and linear programs

Let $\mathcal{G} \subset [p_\mathrm{min}, p_\mathrm{max}]$ be a prescribed finite power grid. A configuration $c = (u, v, s, p_u, p_v)$ includes its resource, pair and two powers. Let $\mathcal{C}_\mathcal{G}$ contain all such configurations, $a_{ic} = 1$ when user i belongs to c, $b_{sc} = 1$ when c uses resource s, $d_c = p_u + p_v$, and $q_c = q_{uvs}(p_u, p_v)$. The grid integer program (IP) is

$$\begin{aligned}
\mathcal{P}_\mathrm{IP}(\mathcal{G}): \quad \max_{z} \quad & \sum_{c} q_c z_c \\
\text{s.t.} \quad & \sum_{c} a_{ic} z_c = 1, \quad i \in \mathcal{K}, \\
& \sum_{c} b_{sc} z_c = 1, \quad s \in \mathcal{S}, \\
& \sum_{c} d_c z_c \le P, \qquad z_c \in \{0, 1\}
\end{aligned} \tag{7}$$

The grid linear program (LP), $\mathcal{P}_\mathrm{LP}(\mathcal{G})$, replaces integrality with $z_c \ge 0$. The incidence equalities imply $z_c \le 1$. Its dual uses unrestricted user prices α_i, unrestricted resource prices β_s, and a nonnegative power price λ:

$$\begin{aligned} \min_{\alpha,\beta,\lambda} \quad & \sum_i \alpha_i + \sum_s \beta_s + \lambda P \\ \text{s.t.} \quad & \alpha_u + \alpha_v + \beta_s + \lambda(p_u + p_v) \geq q_{uvs}(p_u, p_v), \\ & c \in \mathcal{C}_{\mathcal{G}}, \\ & \lambda \geq 0 \end{aligned} \tag{8}$$

The grid LP upper-bounds the grid IP, but may lie below the continuous optimum in (6): powers omitted from $\mathcal{G}$ can yield better allocations. A feasible grid IP schedule, in contrast, is feasible for (6) and supplies a lower bound after its rate is validated. Thus the grid gives useful candidates without by itself certifying the continuous optimum.

Replacing the finite grid by each pair's complete power rectangle gives a configuration relaxation in which nonnegative measures can mix power configurations. Its dual imposes (8) for every power in the rectangle. Denote its value by V_{CLP} and the optimum of (6) by V_{C}. The relaxation permits both assignment mixtures and power mixtures, so in general $V_{\mathrm{C}} \leq V_{\mathrm{CLP}}$ with a nonzero gap.

IV. Structural Analysis

A. Hardness with a fixed receiver dimension

We first ask whether a schedule can attain the interference-free upper limit. In the construction below, attaining that limit forces every selected pair to have orthogonal channels. This connects a rate threshold to a discrete matching decision.

Theorem 1. Fixed-power ideal-threshold attainment for (6) is NP-complete even with $M = 3$, unit-norm real rational channels drawn from a fixed finite alphabet, unit powers and unit white-noise variance. Maximizing the sum rate is therefore NP-hard. For any fixed rational constants $0 < p_{\min} < \bar{p} < p_{\max}$, the corresponding continuous-power family with $P = K\bar{p}$ is also NP-hard through ideal-threshold attainment. The construction has a positive constant additive separation, in natural-log sum-rate units, between yes and no instances.

Proof sketch. Reduce bounded-degree perfect three-dimensional matching [29] to users $X \cup Y$ and resources Z. Each resource's admissible graph has at most three edges and is a forest. Supplementary Section S1 gives a fixed rational unit-vector alphabet in $\mathbb{R}^3$ whose zero inner products realize exactly these edges, including an outsider role. At unit powers a pair reaches $2\ln 2$ exactly when orthogonal, so the network reaches $K \ln 2$ exactly for a perfect matching. The decision is checked by the rational product $\prod_i (1+\gamma_i) \geq 2^K$. For continuous powers, Jensen's inequality bounds the interference-free part by $K\ln(1+\bar{p})$; every forbidden pair incurs a positive loss bounded below using $p_{\min}$ and the smallest nonzero alphabet inner product. Section S11 gives the full reduction and additive separation. □

The contribution is the conjunction of restrictions in Theorem 1. The channel-orthogonality and fixed-antenna precedents discussed in Section II remain essential context.

Theorem 1 does not establish a constant relative approximation gap, nor NP-membership for unrestricted continuous thresholds. The dimension three is sufficient, not a proved boundary between tractable and intractable cases. Its channels can vary independently across resources; it does not assert that the construction can be realized by a prescribed common-ray propagation geometry. Fixed-dimensional representations of forests have established antecedents [30]; the theorem concerns their restricted scheduling consequence.

B. Sharp pairwise curvature

Theorem 2. The function q_r in (4) is jointly concave over $\mathbb{R}^2_+$ if and only if

$$r \leq 1 - \frac{1}{\sqrt{2}}. \tag{9}$$

Within this range its Hessian is negative definite at every finite point.

Proof sketch. Let $a = 1 - r$. At $x = 0$, q_{xx} tends to $1 - 2a^2$ as $y \to \infty$, proving necessity by continuity to an interior point. For $a \geq 1/\sqrt{2}$, the Hessian diagonals are nonpositive. Its determinant has a polynomial numerator with nonnegative coefficients on this range and a strictly positive constant term. Thus the Hessian is negative definite. Sections S2 and S11 provide the coefficients and full algebra. □

This squared-correlation threshold applies to the whole nonnegative quadrant. A finite rectangle can remain concave beyond it; failure of global concavity does not imply positive curvature at every operating point.

C. Complete rectangular power pricing

To extend a configuration dual from grid points to continuous powers, we must bound the rate minus its power cost over every pair rectangle. We call this maximization power pricing:

$$\begin{aligned} H = \max_{(x,y)\in\mathcal{B}} \{q_r(x,y) - \lambda_1 x - \lambda_2 y\}, \\ \mathcal{B} = [L_x, U_x] \times [L_y, U_y]. \end{aligned} \tag{10}$$

For positive gains and a network price λ in natural-log rate units, $\lambda_1 = \lambda/g_{us}$ and $\lambda_2 = \lambda/g_{vs}$. The following result also allows signed prices, although the network power multiplier is nonnegative.

Theorem 3. For $0 < r \leq 1$ and a compact nondegenerate rectangle in $\mathbb{R}^2_+$, the maximum in (10) occurs

among at most seventeen points: four corners, at most two stationary points on each of four edges, and at most five interior stationary points.

Proof sketch. Multiplying the first interior stationarity equation by its positive denominator gives $C(x) + B(x)y = 0$. For $r > 0$, $B = C = 0$ is impossible. Substituting $y = -C/B$ into the second equation yields a polynomial $P(x)$ of degree at most five with $P(-1) = 2r^3 > 0$; hence it is not identically zero and supplies at most five interior candidates. Each edge has a nonzero stationary polynomial of degree at most two. Together with four corners, compactness gives at most seventeen candidates. The explicit polynomials and admissibility checks are in Section S11. □

When $r = 0$, the objective is separable. In coordinate i, use $\mathrm{clip}(1/\lambda_i - 1, [L_i, U_i])$ if $\lambda_i > 0$, and U_i otherwise. A collapsed rectangle is a one-dimensional edge problem or a singleton. In physical coordinates, a zero-gain user contributes only $-\lambda p_i$, maximized at minimum power for the nonnegative network price; the other coordinate is a scalar logarithmic problem. This handles zero channels without dividing by a zero gain. The result does not cover an additional coupled pair constraint without examining its boundary.

Theorem 3 gives finitely many candidates, not a radical formula or precision-independent runtime. Certification requires all relevant real roots and directed logarithmic enclosures; unresolved enclosures are failures. The objective distinguishes this result from earlier elimination methods [13].

V. Certified Network Optimization

A. Repairing a numerical configuration dual

Numerically computed prices may satisfy only the configurations present in a grid master. The following repair raises each resource price enough to cover every continuous configuration, producing a valid bound even when those numerical prices are not optimal.

Proposition 1. Let α_i be arbitrary real values and $\lambda \geq 0$. Suppose H_{uvs} is a valid upper bound on

$$\max_{p_u, p_v \in [p_{\min}, p_{\max}]} \{q_{uvs}(p_u, p_v) - \lambda(p_u + p_v)\}. \tag{11}$$

Set $\beta_s = \max_{u<v}(H_{uvs} - \alpha_u - \alpha_v)$. Then

$$\begin{aligned} U_{\mathrm{dual}} &= \sum_i \alpha_i + \sum_s \beta_s + \lambda P, \\ V_{\mathrm{C}} &\leq V_{\mathrm{CLP}} \leq U_{\mathrm{dual}}. \end{aligned} \tag{12}$$

Proof. The choice of β_s makes every continuous configuration inequality in (8) valid. Multiply those inequalities by any feasible configuration mass and sum. User and resource equalities produce the first two sums in (12), while nonnegativity of λ bounds the power contribution by λP. This is weak duality and does not require that the candidate prices solve the numerical master exactly. □

Every pair/resource rectangle must be covered, including pairs absent from the numerical master. An omitted rectangle needs a separate valid bound. Evaluating a feasible integral schedule with downward rounding gives $L \leq V_{\mathrm{C}}$. Together with a validated repair, this yields $L \leq V_{\mathrm{C}} \leq U_{\mathrm{dual}}$, so the unknown loss $V_{\mathrm{C}} - L$ is at most $U_{\mathrm{dual}} - L$. A valid interval is a certificate of these inequalities; attaining a prescribed gap additionally requires that the interval be sufficiently narrow.

Dual repair does not remove the configuration integrality gap. Supplementary Section S3 gives a rational-channel assignment example and a separate fixed-pair power example. In the latter, $g_1 = g_2 = r = 1$, $p_i \in [1, 2]$ and $P = 11/5$. The deterministic optimum is $\ln(128/55)$, but a configuration mixture with masses $4/5$ at $(1, 1)$ and $1/5$ at $(1, 2)$ has the same average power and strictly larger objective. Therefore branching only on assignments is not in general sufficient for integral-continuous convergence.

B. Complete small-network reference

There are $K!/2^{K/2}$ assignments of unordered user pairs to labelled resources. Enumerating this family gives a complete reference for small networks, but its growth limits scale. For each assignment, the remaining domain is a box in K physical power variables intersected with $\sum_i p_i \leq P$. Subdividing these powers, rather than only selecting assignments, allows both assignment and power-mixing gaps to be resolved.

For a normalized pair box $[\ell_x, u_x] \times [\ell_y, u_y]$, let $a = 1 - r$. The mixed-monotone bound is

$$U_{\mathrm{MM}} = \ln\left(1 + u_x \frac{1 + a\ell_y}{1 + \ell_y}\right) + \ln\left(1 + u_y \frac{1 + a\ell_x}{1 + \ell_x}\right). \tag{13}$$

Each term increases in its own power and decreases in the interfering normalized power, proving the bound directly from (3). A tighter budget-aware alternative follows by decomposing $q = f + g_x + g_y$, where $f = 2 \ln d$ and $g_x = -\ln(1 + x)$. The Hessian of f is

$$\nabla^2 f = -\frac{2}{d^2} \begin{bmatrix} (1 + ay)^2 & r \\ r & (1 + ax)^2 \end{bmatrix}. \tag{14}$$

The matrix inside (14) is positive semidefinite because $(1 + ax)(1 + ay) \geq 1 \geq r$. Hence f is concave and its tangent at any box point majorizes it. Each g_i is convex and lies below the chord between its two endpoints. Adding the tangent and both chords gives an affine upper bound on the whole pair box. Degenerate intervals use their point value.

Summing these affine bounds over the assignment and maximizing over the physical power box and total budget is a continuous knapsack problem: start at the lower powers, allocate remaining budget to positive affine slopes in descending order, and stop at each upper bound. The resulting upper bound retains the total-power constraint, unlike independently maximizing every pair. Taking the minimum of it and the summed bounds (13) preserves validity. The tangents, secants and knapsack arithmetic must be evaluated with outward bounds.

When the exact rational condition $(1 - r)^2 \geq 1/2$ holds, Theorem 2 supplies a supporting tangent for the entire pair rate. The accelerated reference uses this tighter affine bound in place of the tangent-secant construction for that pair, retaining the same budget maximization and mixed-monotone intersection. For all remaining pairs it retains the generic bound.

The reference first covers every assignment with a valid upper bound. It then selects a live box with largest bound and bisects a longest physical-power interval. Boxes whose lower powers violate the budget are infeasible; boxes whose upper bounds cannot improve the incumbent are discarded. The lower corner of every retained box is feasible, and additional feasible points may improve the incumbent. At interruption, the maximum live upper bound, combined with the incumbent, remains a valid global interval. The separately computed U_{dual} may be intersected with this upper bound without changing feasibility or completeness.

Proposition 2. In exact arithmetic, with finitely many assignments and the stated compact power domain, exhaustive longest-edge subdivision with continuous, singleton-consistent bounds converges to the global value. For any positive requested tolerance, the reference terminates after finitely many subdivisions if computation is unrestricted and all potentially improving boxes are eventually refined.

Proof. The objective is uniformly continuous on each compact feasible power domain. Both the mixed-monotone and tangent-secant bounds agree with the objective at singleton boxes, and their errors vanish as box diameter tends to zero. Longest-edge bisection forces diameters to zero along every infinite refinement branch. Finite assignment coverage and exhaustive refinement therefore rule out a persistent upper-bound excess greater than the requested tolerance. Feasible lower corners approach any feasible optimizer as its containing boxes shrink. □

Proposition 2 establishes convergence, not scalable complexity. Finite precision imposes an arithmetic floor, which must be reported if it prevents the requested enclosure. The pair oracle does not remove scheduling hardness or establish a speedup. The evaluated implementation prices complete rectangular partitions; the algebraic oracle is examined separately.

TABLE II
CERTIFIED OPTIMIZATION PROCEDURE

1. Construct a feasible uniform schedule and a valid interference-free ceiling.
2. Solve the numerical grid IP/LP; validate any improved integral incumbent with outward rate bounds.
3. In the full arm, allocate at most one quarter of the supplied continuation time to price all pair/resource rectangles and repair the dual. Incomplete pricing retains the valid fallback.
4. If time remains, cover every labelled assignment with spatial bounds, adding pair tangents where Theorem 2 applies. Retain the fallback until coverage is complete.
5. Refine the live box with largest bound; stop refining infeasible or dominated boxes while retaining proof coverage. At interruption retain the timely incumbent and a global upper bound, intersected with any complete repaired dual.

A separate checker reconstructs gains and correlations from stored binary channels, validates schedule/power constraints, and encloses logarithms with integer arithmetic and explicit remainders. Exported partition trees and dual weights establish complete coverage; unvisited assignments retain the interference-free fallback. Recorded verification times determine deadline attainment independently of later worker returns. The supplement describes replay and reuse of identical inputs. Exactness concerns the stored nominal channels.

C. Evaluation quantities and comparison contract

We report validated feasible values $L/\ln 2$, global gaps $U - L$ in nat, radiated power, deadline attainment and failures. Improving a feasible value increases achieved rate; reducing an upper bound improves knowledge of allocation quality. These are separate outcomes. Comparators use the same channels, receiver, service constraints, power domain and scheduling family, with schedule search included in their computational allowance.

VI. NUMERICAL RESULTS

A. Physical inputs and comparison protocol

We use DeepMIMO's geometry-based channel generation [31] to evaluate complete schedules in two ray-tracing scenes: O1_60 at 60 GHz and ASU_Campus1 at 3.5 GHz. Each scene supplies one eight-antenna access point with linear-MMSE reception. Channels retain up to ten propagation paths and their path loss. We assume ideal synchronization, block-static channels and no inter-resource interference or estimation error. The two bands provide contrasting propagation examples; they do not constitute standardized sixth-generation deployment models.

The results below come from a separately registered correction to the original study. Its 32-point, 50-MHz

delay window had clipped every retained ASU path, producing zero channels. The original outcomes, including the null primary comparison, remain in the supplement. Before generating new outcomes, we specified a 512-point transform, 32 tones at indices $0, 16, \ldots, 496$, and a 512-sample cyclic prefix (CP). The useful symbol and CP each last $10.24\,\mu$s, and all retained absolute delays fit within the guard. The generator normalization is removed by $\sqrt{512}$. Noise at -174 dBm/Hz with a 7-dB noise figure gives 1.9485×10^{-15} W per tone. Users are neither delay-shifted individually nor filtered by channel strength.

We report gross occupied-tone spectral efficiency (SE), Q/S. For a validated lower bound L in natural-log sum-rate units, $\mathrm{SE} = L/(S \ln 2)$. The assumed CP would halve payload SE before any other overhead. Accordingly, these values are neither net SE nor throughput over the full 50-MHz band. The guard is an idealized waveform assumption; we do not perform a time-domain validation.

The corrective pools use previously untouched O1 rows 321–340 and ASU rows 21–40. Each scene supplies 12 eight-user main cases ($S = 4$) and four sixteen-user scale cases ($S = 8$). Fixed seeds select distinct users within a case, with replacement across cases from the same scene pool. All 32 cases are nontrivial, although individually weak or zero channels remain. Powers lie in $[0.05, 0.2]$ W with total budget $K/10$ W. The computational budget determined sample sizes; no prospective statistical power calculation was used. Inference is conditional on these two scenes.

Table III separates each method's feasible search from its upper bound. Grid uses $\{0.05, 0.1, 0.2\}$ W; its integer linear program (ILP) and LP relaxation alone do not bound the continuous optimum. Matching–WMMSE, labelled Joint in compact plots and tables, combines resource-labelled matching, receiver/weight updates and shared-budget amplitude updates, with nine starts and bounded exchange refinement. Its scheduling step draws on Shen and Yu [11], and its weighted-MMSE updates on Wang et al. [12]. Both are adapted to the present full-service model, so the comparator does not reproduce either source's full system or reported curves. Eight-user runs receive 18 s with a 20-s external watchdog; scale runs receive 28 s with a 30-s watchdog. The allowance includes startup, search, separate certificate checking and persistence of the verified result. Common finite iteration and subdivision caps can cause normal return before the time limit. Full uses validated rectangular subdivision for pricing; the seventeen-candidate characterization is examined separately and does not explain the measured network runtimes. Runs are serial on the recorded nondedicated Windows host, with single-threaded numerical linear algebra.

TABLE III
COMMON-MODEL METHODS. ALL USE THE SAME INPUTS, CONSTRAINTS AND CHARGED ALLOWANCE.

Arm	Feasible search	Continuous upper bound
Grid	Grid ILP and LP	Interference-free ceiling
Curvature	Grid; complete scheduling family; bounded power refinement	Spatial plus eligible pair tangents
Full	Same search; additional box pricing	Spatial/dual intersection
Matching–WMMSE	Alternating matching and continuous powers; exchanges	Common ceiling
Grid + pricing	Grid incumbent, at $K = 16$	Repaired dual after all prices

The registered primary endpoint compares full with curvature: the equal-scene-weighted difference in the fraction of runs that return normally with a verified gap at most 0.01 nat by the charged deadline. Four 98.75% Clopper–Pearson intervals for scene-specific discordant probabilities give a conservative at-least-95% Bonferroni interval for that difference. Secondary paired-bootstrap intervals use 10,000 fixed-protocol resamples and are exploratory. Failures and missing results remain in their denominators. Original and corrective outcomes are analyzed separately; the supplement gives both protocols, the eight analytic reliability executions, mechanism controls and postprocessing deviations.

B. Global bounds and execution reliability

All 128 corrective executions returned normally, without recorded solver or storage errors or verification timeouts. A separate offline checker reconstructed the accepted bounds from the saved evidence. Automatic postprocessing exceeded its initial allowance, so one disclosed 92.89-s saved-data reproduction completed the analysis and replay without rerunning a performance cell. This extra cost is separate from the charged method durations. The supplement reports all 700 receipts and the 417 checks that reused identical mathematical inputs after individual provenance checks.

Fig. 3 shows all 12 main cases per scene. Every method returns normally, but none reaches the 0.01-nat target. Full and curvature therefore have zero successes and 12 joint failures in each scene, giving a primary difference of zero with conservative 95% interval $[-0.3449, 0.3449]$. This establishes neither superiority nor equivalence. Normal completion is observed for these executions; useful target certification is not.

Fig. 4 shows the empirical cumulative distribution of verified global gaps: higher curves at a fixed horizontal value mean more cases with no larger gap. The loga-

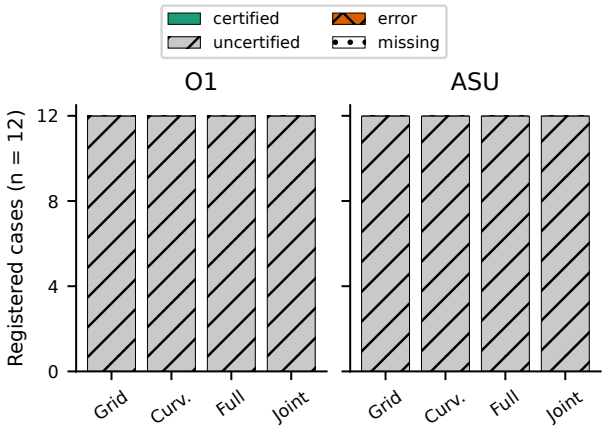


Fig. 3. Operational target attainment: 12 corrective eight-user cases per scene and arm, 18-s allowance. All are normal but uncertified; zero-height error and missing categories are retained.

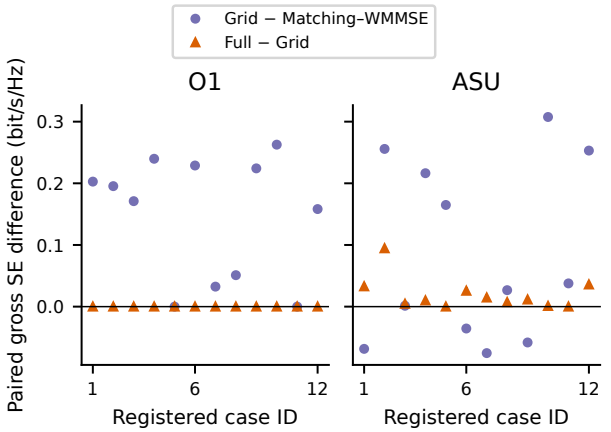


Fig. 5. Paired gross occupied-tone SE differences in the 24 corrective main cases. Zero denotes equal achieved rates. Case IDs are fixed identifiers, not a time or parameter sweep. Full and curvature coincide.

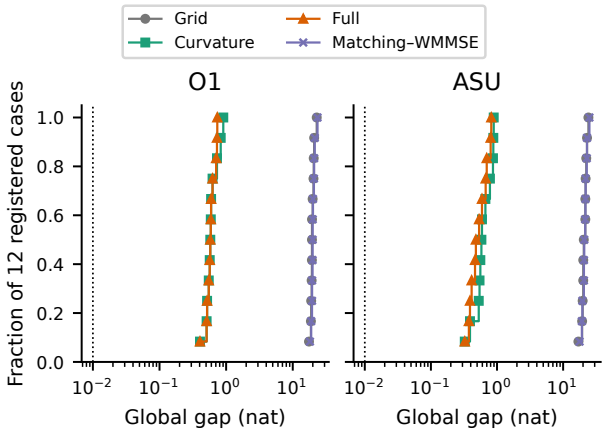


Fig. 4. Continuous global-gap distributions with all 12 cases per curve. The dotted line marks 0.01 nat. Leftward shifts indicate tighter bounds, not higher achieved rates.

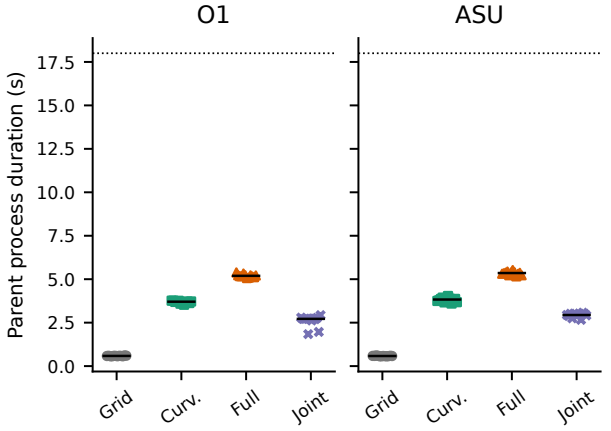


Fig. 6. All main-case parent durations; black bars show medians, not confidence intervals. The dotted line is the 18-s charged cap. Every observed return is normal. Measurements concern one nondedicated host.

rithmic axis separates the grid ceiling from tighter continuous bounds. Mean gaps for grid, curvature and full are respectively 19.784, 0.617 and 0.592 nat in O1, and 20.970, 0.634 and 0.547 nat in ASU. The equal-scene full-minus-curvature mean is -0.0554 nat, with exploratory paired-bootstrap interval $[-0.0851, -0.0295]$. Pricing reduces some upper bounds without creating a target certificate. These gaps bound unknown regret; they are not measured optimality errors.

Fig. 5 separates achieved-rate gains from bound tightening. Grid exceeds matching–WMMSE by a mean 0.1472 bit/s/Hz in O1 and 0.0855 in ASU. Full and curvature achieve identical rates: both equal grid in O1, while their mean ASU improvement over grid is 0.0198 bit/s/Hz. The extra ASU rate therefore comes from their shared refinement. The equal-scene full-minus-matching–WMMSE mean is 0.1263 bit/s/Hz, with exploratory interval $[0.0766, 0.1768]$. This comparison concerns the stated implementations, inputs and budgets; it does not rank the broader methods across other models.

Fig. 6 reports parent-process duration, including startup and checking. The full arm averages 5.20 and 5.34 s in O1 and ASU, versus 3.68 and 3.81 s for curvature. Grid averages about 0.58 s and matching–WMMSE 2.60 and 2.93 s. Pricing therefore adds cost in this implementation. The methods stop at their declared finite work caps, well before the 18-s ceiling; these are not time-to-convergence or time-to-target measurements. Early normal return should not be interpreted as a real-time guarantee.

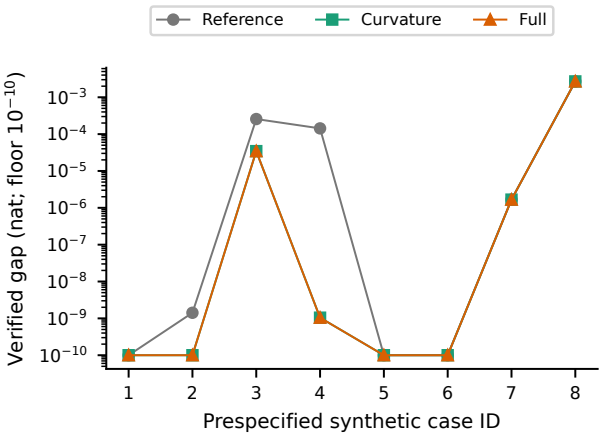


Fig. 7. Same-incumbent mechanism controls, eight original-study synthetic cases and 2-s searches plus verification. The logarithmic display floor is 10^{-10} nat. These are separate from the corrective ray-traced primary population.

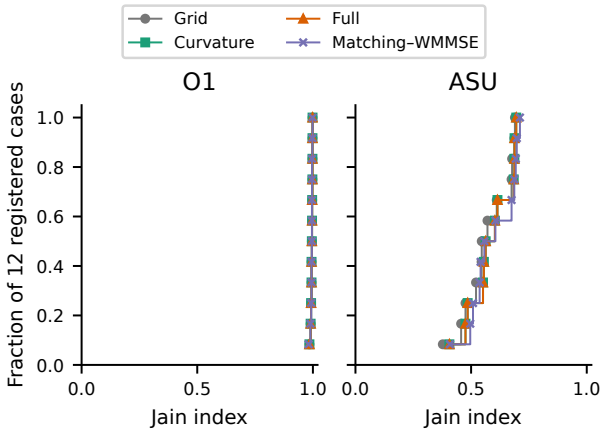


Fig. 8. Jain-index distributions over 12 cases per scene, normalized by that registered count. Undefined all-zero values would remain missing. Fairness is descriptive, not the optimization objective.

C. Controlled mechanism and fairness

Fig. 7 uses the eight prespecified synthetic four-user cases from the original study, reused once rather than regenerated. Reference, curvature and full continuations begin from the same incumbent and dual, with equal 2-s search caps followed by separate verification. All meet the target; eligible tangents tighten several low-alignment, high-gain cases. Full's repaired dual never improves the final spatial bound in these eight cases. The figure therefore isolates a curvature mechanism but does not demonstrate an incremental pricing benefit. Values below 10^{-10} nat share a display floor; they are enclosures, not exact zero errors. The supplement specifies all gain, alignment and seed combinations.

Fig. 8 compares user-rate fairness. Higher Jain indices indicate more balanced rates, but sum-rate maximization imposes no fairness objective. Full has mean index 0.9947 in O1 and 0.5840 in ASU, while matching–WMMSE gives 0.9940 and 0.5930. ASU thus exposes a rate/fairness tradeoff rather than a uniform benefit. Full's mean minimum-user SE is 0.1530 bit/s/Hz in ASU, versus 0.1958 for matching–WMMSE, although both serve every user with positive power. Positive power and full service do not ensure comparable rates for weak users.

D. Sixteen-user results and limits

Fig. 9 retains all four scale cases in each scene. Grid and grid-plus-pricing have identical rates, averaging 27.827 bit/s/Hz in O1 and 9.598 in ASU; matching–WMMSE averages 27.495 and 9.423. The grid accounts for the observed rate advantage. The scale pricing arm avoids enumerating the complete scheduling family, so it

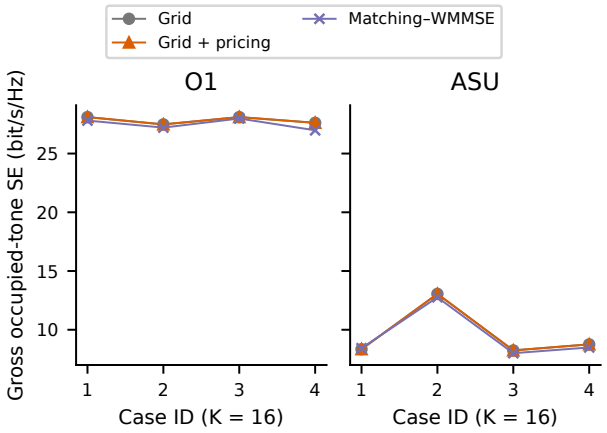


Fig. 9. Gross occupied-tone SE for all eight sixteen-user cases, 28-s allowance per arm. Grid and pricing overlap. Lines connect arbitrary case IDs solely for readability.

is distinct from the eight-user spatial method. Four cases per scene do not establish a scaling law.

Fig. 10 shows relative global gaps, $100(U-L)/L$ for $L > 0$; smaller bars mean less uncertainty about the optimum. Pricing covers all 960 pair/resource rectangles in each scale case. Mean absolute gaps decrease from 38.863 to 7.225 nat in O1 and from 38.580 to 3.933 nat in ASU. Mean relative gaps decrease from 25.20% to 4.68% and from 74.58% to 7.47%, respectively. None reaches 0.01 nat. Thus all eight observed scale bounds improve, while achieved rates remain unchanged and a substantial global gap remains.

VII. Discussion

Pricing tightens bounds without increasing achieved rate or meeting the target. The complete small-network

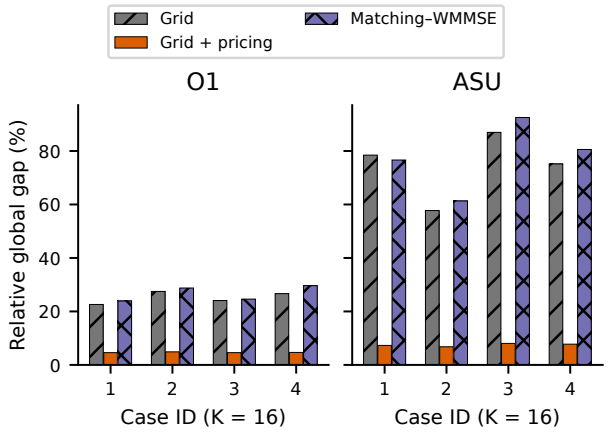


Fig. 10. Relative global gaps in all sixteen-user cases. These percentages bound unknown regret and are not measured rate losses. All pricing runs cover 960/960 rectangles; none meets the absolute target.

reference therefore supports offline assessment, not scalable near-optimal or real-time operation.

Separate CSI and GPU studies (Supplementary Sections S6 and S5) limit broader claims. The 120-network, four-user CSI study establishes neither CNN superiority nor equivalence at four pilots; its normalized units differ from the physical-watt study. GPU coefficient construction is faster only at the largest tested size, while GPU LP optimization shows no speedup and produces infeasible iterates at 16 and 32 users. These findings support no learned-CSI, robustness or end-to-end GPU acceleration guarantee.

VIII. Conclusion

Full-service MMSE scheduling remains hard with three receive dimensions and a finite rational channel alphabet. Sharp concavity and finite rectangular pricing support continuous bounds, while assignment and power mixing explain why the configuration relaxation need not be exact.

Pricing tightens 11 of 24 main upper bounds and all eight scale bounds, without additional rate over the corresponding search without pricing. Separate replay reconstructs the accepted bounds. No corrective main or scale case meets 0.01 nat, and the registered comparison establishes neither superiority nor equivalence. Further work should improve bound refinement, test broader propagation populations and develop channel-uncertainty models.

Acknowledgment

OpenAI Codex (GPT-6-family models, including Astra) assisted mathematical derivations and checks, experimental and analysis code, literature synthesis, and drafting throughout all main and supplementary sections. Tables and plots use recorded outputs and AI-assisted code; accompanying evidence documents validation limits.

Data Availability

Code, configurations, generated results, verification receipts and replay instructions accompany the paper (Supplementary Section S10). Raw DeepMIMO scenarios are excluded; public access to the revised project repository release remains pending verification.

References

[1] T. Haustein, "Real time signal processing for multi-antenna systems and experimental verification on a reconfigurable hardware test-bed," Ph.D. dissertation, Technische Universität Berlin, 2006. [Online]. Available: https://edocs.tib.eu/files/e01dd01/522593321.pdf

[2] J. Fan, G. Y. Li, Q. Yin, B. Peng, and X. Zhu, "Joint user pairing and resource allocation for LTE uplink transmission," *IEEE Trans. Wireless Commun.*, vol. 11, no. 8, pp. 2838–2847, 2012, doi: 10.1109/TWC.2012.070912.111305.

[3] M. A. Ruder, A. Moldovan, and W. H. Gerstacker, "SC-FDMA user pairing and frequency allocation with imperfect channel state information," *Phys. Commun.*, vol. 13, pp. 88–98, 2014, doi: 10.1016/j.phycom.2014.10.001.

[4] M. A. Ruder, D. Ding, U. L. Dang, A. V. Vasilakos, and W. H. Gerstacker, "Joint user grouping and frequency allocation for multiuser SC-FDMA transmission," *Phys. Commun.*, vol. 8, pp. 91–103, 2013, doi: 10.1016/j.phycom.2013.02.004.

[5] N. Prasad, H. Zhang, H. Zhu, and S. Rangarajan, "Multi-User MIMO Scheduling in the Fourth Generation Cellular Uplink," *IEEE Trans. Wireless Commun.*, vol. 12, no. 9, pp. 4272–4285, 2013, doi: 10.1109/twc.2013.072513.120734.

[6] L. Salaün, C. S. Chen, and M. Coupechoux, "Optimal Joint Subcarrier and Power Allocation in NOMA Is Strongly NP-Hard," in *2018 IEEE International Conference on Communications (ICC)*, 2018, pp. 1–7, doi: 10.1109/icc.2018.8422362.

[7] M. Mollanoori and M. Ghaderi, "On the complexity of wireless uplink scheduling with successive interference cancellation," in *Proc. 49th Allerton Conference on Communication, Control, and Computing*, 2011, pp. 1064–1071, doi: 10.1109/allerton.2011.6120285.

[8] M. Razaviyayn, M. Sanjabi, and Z.-Q. Luo, "Linear Transceiver Design for Interference Alignment: Complexity and Computation," *IEEE Trans. Inf. Theory*, vol. 58, no. 5, pp. 2896–2910, 2012, doi: 10.1109/tit.2012.2184909.

[9] Y.-F. Liu, "Dynamic Spectrum Management: A Complete Complexity Characterization," *IEEE Trans. Inf. Theory*, vol. 63, no. 1, pp. 392–403, 2017, doi: 10.1109/tit.2016.2623662.

[10] K. Shen and W. Yu, "Fractional Programming for Communication Systems—Part I: Power Control and Beamforming," *IEEE Trans. Signal Process.*, vol. 66, no. 10, pp. 2616–2630, 2018, doi: 10.1109/tsp.2018.2812733.

[11] ——, "Fractional Programming for Communication Systems—Part II: Uplink Scheduling via Matching," *IEEE Trans. Signal Process.*, vol. 66, no. 10, pp. 2631–2644, 2018, doi: 10.1109/tsp.2018.2812748.

[12] Z. Wang, J. Zhang, H. Q. Ngo, B. Ai, and M. Debbah, "Uplink Precoding Design for Cell-Free Massive MIMO With Iteratively Weighted MMSE," *IEEE Trans. Commun.*, vol. 71, no. 3, pp. 1646–1664, 2023, doi: 10.1109/tcomm.2023.3235919.

[13] N. Ul Hassan, C. Yuen, S. Saeed, and Z. Zhang, "Power Control for Sum-Rate Maximization on Interference Channels Under Sum Power Constraint," *IEEE Trans. Veh. Technol.*, vol. 64, no. 2, pp. 593–609, 2015, doi: 10.1109/tvt.2014.2324289.

[14] S. Sharma and D. Teneketzis, "An externalities-based decentralized optimal power allocation algorithm for wireless networks," *IEEE/ACM Trans. Netw.*, vol. 17, no. 6, pp. 1819–1831, 2009, doi: 10.1109/TNET.2009.2020162.

[15] B. Matthiesen, C. Hellings, E. A. Jorswieck, and W. Utschick, "Mixed Monotonic Programming for Fast Global Optimization," *IEEE Trans. Signal Process.*, vol. 68, pp. 2529–2544, 2020, doi: 10.1109/tsp.2020.2983284.

[16] C. Barnhart, E. L. Johnson, G. L. Nemhauser, M. W. P. Savelsbergh, and P. H. Vance, "Branch-and-price: Column generation for solving huge integer programs," *Oper. Res.*, vol. 46, no. 3, pp. 316–329, 1998, doi: 10.1287/opre.46.3.316.

[17] Y. Zhao and G. J. Pottie, "Optimal Spectrum Management in Multiuser Interference Channels," *IEEE Trans. Inf. Theory*, vol. 59, no. 8, pp. 4961–4976, 2013, doi: 10.1109/tit.2013.2255731.

[18] S. Mostafa, C. W. Sung, and Y. Guo, "Joint computation and communication resource allocation with NOMA and OMA offloading for multi-server systems in F-RAN," *IEEE Access*, vol. 10, pp. 24 456–24 466, 2022, doi: 10.1109/ACCESS.2022.3152531.

[19] S. Shi, M. Schubert, and H. Boche, "Rate optimization for multiuser MIMO systems with linear processing," *IEEE Trans. Signal Process.*, vol. 56, no. 8, pp. 4020–4030, 2008, doi: 10.1109/TSP.2008.924796.

[20] E. Viterbo and A. Hottinen, "Optimal user pairing for multiuser MIMO," in *Proc. IEEE ISSSTA*, 2008, pp. 242–246, doi: 10.1109/ISSSTA.2008.50.

[21] X. Ou, S. Dang, and A. Doufexi, "Uplink RSMA with dual-polarized multi-user MIMO over correlated rician channels in finite blocklength regime," *IEEE Trans. Commun.*, vol. 74, pp. 14 516–14 531, 2026, doi: 10.1109/TCOMM.2026.3731763.

[22] K. Chen, C. Qi, and O. A. Dobre, "Integrated User Grouping, Subcarrier Allocation, and Hybrid Beamforming for Wideband Multiuser mmWave Massive MIMO," *IEEE Trans. Commun.*, vol. 74, pp. 3313–3328, 2026, doi: 10.1109/TCOMM.2026.3652527.

[23] M. Elyasi and A. Vosoughi, "Massive MIMO Over Correlated Fading Channels: Multi-Cell MMSE Processing, Pilot Assignment and Power Control," *IEEE Trans. Wireless Commun.*, vol. 25, pp. 30–46, 2026, doi: 10.1109/TWC.2025.3581057.

[24] S. Daei, G. Fodor, M. Skoglund, and M. Telek, "Toward Optimal Pilot Spacing and Power Control in Multi-Antenna Systems Operating Over Non-Stationary Rician Aging Channels," *IEEE Trans. Commun.*, vol. 73, no. 6, pp. 3761–3777, 2025, doi: 10.1109/TCOMM.2024.3490500.

[25] J. Fu, P. Jiang, P. Zhu, H. Q. Ngo, and M. Matthaiou, "Deterministic equivalent-based resource allocation for cell-free massive MIMO," *IEEE Trans. Commun.*, vol. 74, pp. 6443–6460, 2026, doi: 10.1109/TCOMM.2026.3672179.

[26] Z. Liu, J. Zhang, Y. Zhu, E. Shi, and B. Ai, "Mobile Cell-Free Massive MIMO With Multi-Agent Reinforcement Learning: A Scalable Framework," *IEEE Trans. Wireless Commun.*, vol. 24, no. 3, pp. 1817–1831, 2025, doi: 10.1109/TWC.2024.3512663.

[27] Z. Mobini, A. H. Gokceoglu, L. Wang, G. Peters, and H. Q. Ngo, "Fronthaul-Aware User-Centric Generalized Cell-Free Massive MIMO Systems," *IEEE Trans. Wireless Commun.*, vol. 25, pp. 900–914, 2026, doi: 10.1109/TWC.2025.3587672.

[28] J. Edmonds, "Maximum matching and a polyhedron with 0,1-vertices," *Journal of Research of the National Bureau of Standards, Section B: Mathematics and Mathematical Physics*, vol. 69B, no. 1–2, pp. 125–130, 1965, doi: 10.6028/jres.069b.013.

[29] M. E. Dyer and A. M. Frieze, "On the complexity of partitioning graphs into connected subgraphs," *Discrete Appl. Math.*, vol. 10, no. 2, pp. 139–153, 1985, doi: 10.1016/0166-218x(85)90008-3.

[30] AIM Minimum Rank–Special Graphs Work Group, "Zero forcing sets and the minimum rank of graphs," *Linear Algebra Appl.*, vol. 428, no. 7, pp. 1628–1648, 2008, doi: 10.1016/j.laa.2007.10.009.

[31] A. Alkhateeb, "DeepMIMO: A generic deep learning dataset for millimeter wave and massive MIMO applications," in *Proc. Information Theory and Applications Workshop (ITA)*, San Diego, CA, USA, Feb. 2019, pp. 1–8. [Online]. Available: https://www.wi-lab.net/research/localization-digital-twins/